

MECHANICAL DESIGN AND IMPLEMENTATION OF A HIGH HARMONIC GENERATION SOURCE AT THE SXP INSTRUMENT

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Abstract

The first experimental technique implemented at the Soft X-ray Port (SXP) of the European XFEL is femtosecond time-resolved X-ray photoelectron spectroscopy (TR-XPES), enabling studies of ultrafast changes in the electronic, chemical, and atomic structure of materials. It offers femtosecond time resolution and enhanced probing depth via higher electron kinetic energies, enabled by the extended photon energy range from 300 eV to 3 keV. To further expand its capabilities, a laser-based High Harmonic Generation (HHG) source is under development. It will generate extreme ultraviolet (XUV) pulses up to 70 eV using a 1030 nm Ytterbium fiber laser amplifier delivering 200 μ J pulse energy at 334 kHz. This will enable more surface-sensitive measurements, high-fidelity probing of shallow core levels, and valence band mapping with high angular resolution. This contribution presents the mechanical design, key technological developments, implementation, and current status of the HHG source at the SXP instrument of the European XFEL.

INTRODUCTION

The main layout of the SXP Scientific Instrument is shown in Fig. 1. The FEL beam, generated by the SA-SE3 undulator system, is focused using a Kirkpatrick–Baez

(KB) X-ray mirror system and guided to the interaction point through the Photon Arrival Monitor (PAM) diagnostic module and the Laser IN-coupling unit (LIN). An Alignment Laser System (ALAS), located upstream of the KB system, enables pre-alignment of all beamline components without the FEL beam [1].

The HHG system is split in two parts: The HHG driving laser and the main HHG system. The former is an Ytterbium fiber laser amplifier (AFS) with a central wavelength of 1030 nm delivering a pulse energy of up to 200 μ J at 334 kHz. The 300 fs long pulses are compressed down to about 40 fs using a Herriot multi-pass cell and then transported to the main HHG unit. The latter will be placed on a support-table of the (PAM) module [2]. The laser pulses interact non-linearly with Ar gas in a cell at high pressure resulting in high harmonics of the fundamental laser frequency [3-5]. The generated HHG photons are separated from the laser beam and guided into the LIN module via a UHV line. A pair of multilayer mirrors is used to select specific harmonics that are then focused at the interaction point of the TR-XPES experimental station [2]. The FEL and an additional fs optical laser (OL) beam can be colinearly propagated with the HHG beam, allowing to realize different pump-probe schemes beyond the standard OL pump – FEL probe: OL pump – HHG probe, HHG pump – FEL probe, FEL pump – HHG probe.

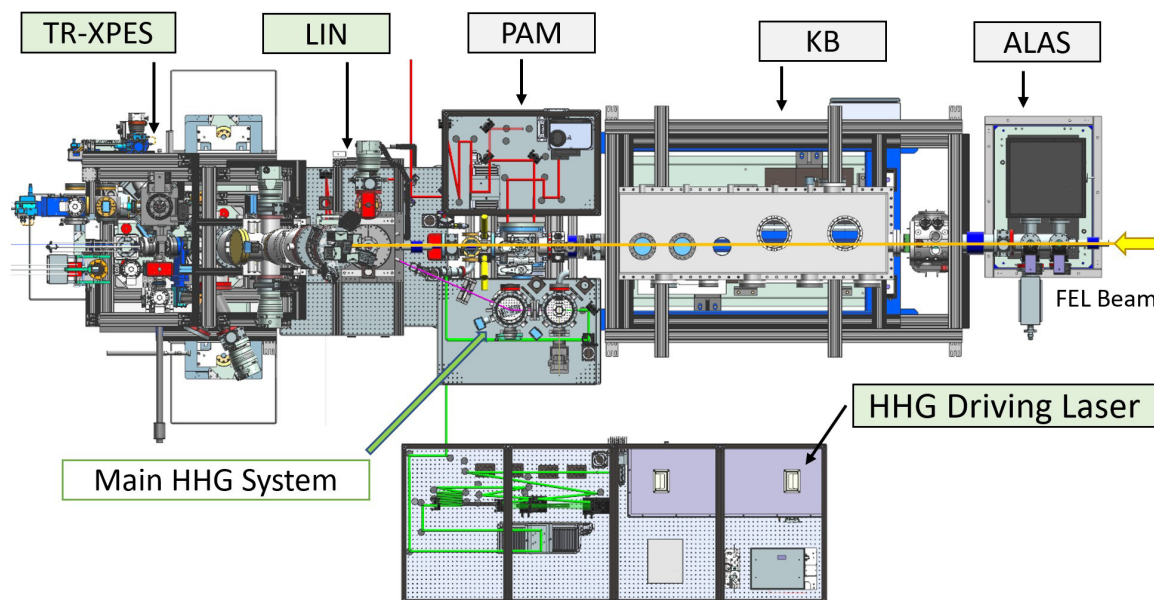


Figure 1: Top view of the SXP instrument, showing all components including the laser-based HHG source. The FEL beam path is indicated in yellow, the laser path in green, and the HHG beam path in purple.

HHG SYSTEM

The main HHG system is shown in Fig. 2. It consists of two vacuum chambers connected via differential pumping. Both vacuum chambers are equipped with individual turbomolecular pumps (Pfeiffer HiPace 700) and Pirani cold cathode vacuum gauges covering the range from atmospheric pressure down to ultrahigh vacuum (UHV).

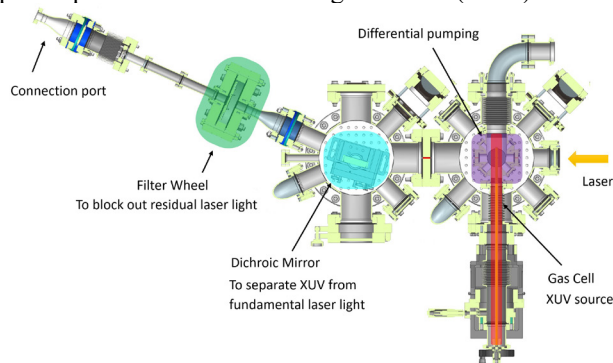


Figure 2: Top view cross-sectional diagram of the HHG system.

The source chamber contains a sub-chamber housing the high-pressure gas cell. Together, they form a two-stage differential pumping system that enables operation of the gas cell at high pressure while maintaining vacuum conditions in the outer chamber by removing residual gases.

The laser beam enters the source chamber through a viewport and is focused into the gas cell sub-chamber through a 1 mm diameter aperture. Within the sub-chamber, the laser interacts with gas molecules inside the integrated gas cell, generating an XUV beam.

The XUV beam is transmitted into a second chamber (the rejector) via a differential pumping section consisting of a CF64 flange with a 3 mm central aperture. In the rejector chamber, the HHG-generated XUV beam is reflected by a dichroic mirror at a 24° grazing incidence angle, allowing reflection of the XUV beam while transmitting a large fraction of the driving laser beam. The reflected beam is directed to the LIN through a UHV beamline equipped with an in-vacuum filter wheel containing various metal films to select the desired XUV wavelength and absorb residual laser light before entering the LIN chamber. In the LIN module, a set of multilayer mirrors is used to select a specific harmonic and focus it at the interaction point of the TR-XPES experimental station.

ROLE OF THE SUB-CHAMBER AND ITS ALIGNMENT WITH GAS CELL

An in-vacuum sub-chamber is integrated into the source chamber to provide differential pumping between the gas cell and the main vacuum chamber. Its primary function is to maintain high-vacuum conditions in the HHG chamber during operation. Since the gas cell operates in continuous mode, the sub-chamber ensures that the pressure in the surrounding chamber remains sufficiently low to minimize XUV reabsorption without overloading the turbopump. A schematic view of the in-vacuum sub-chamber is shown in

Fig. 3. The sub-chamber (6) is connected to the main chamber on both sides via vacuum bellows. On one side, it is connected to the gas cell (1) (Swagelok – 6 mm steel pipe), which is aligned using a 3-degree manipulator (VaB KPM 12-50) with travel ranges of approximately ± 12 mm along each axis.

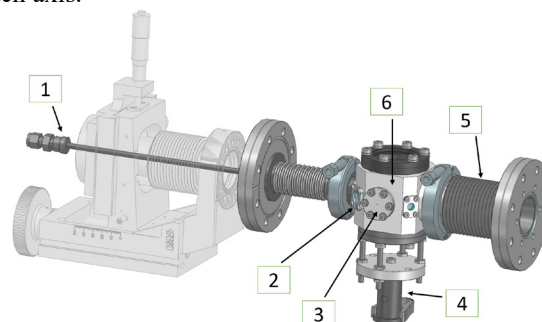


Figure 3: 3D model of the in-vacuum sub-chamber.

The opposite side of the sub-chamber is connected to the main vacuum chamber via an adapter (5), which is linked to a fore-vacuum pump (Pfeiffer – PM 061 356 AT) through a KF50 bellow. The noble gas—initially Argon—is injected through the pipe using a Bronkhorst mass flow controller. Pressure is monitored with a dedicated high-pressure gauge (Inficon – PCG550 Pirani). The sub-chamber is isolated from the main chamber by two small CF16 separation flanges (3), mounted opposite each other and precisely aligned to allow the laser beam to pass through both apertures simultaneously. Additionally, the sub-chamber is equipped with viewports for monitoring the laser–gas interaction region inside the gas cell using dedicated CCD cameras (Basler AG – acA2040-35gm).

The sub-chamber is mounted to the bottom side of the main chamber using an alignment system that provides rigid fixation (4).

Vacuum tests have shown that the source main chamber can be kept around 3×10^{-5} mbar during a continuous gas load of 100 mbar through the gas cell sub-chamber. The alignment of the in-vacuum sub-chamber is performed in two steps. First, the alignment is conducted outside the vacuum chamber by assembling the alignment system, flanges, and viewports onto the sub-chamber, followed by fine adjustment using a green laser diode. In the second step, the sub-chamber is transferred, positioned, and mounted onto the main chamber, establishing connections to the in-vacuum bellows.

Figure 4 shows the alignment scheme for the in-vacuum sub-chamber and the entire HHG section. The alignment procedure uses a green laser diode (Thorlabs PL251). The inset in Fig. 4 displays an image of the alignment laser beam, captured by the detection camera after passing through the sub-chamber.

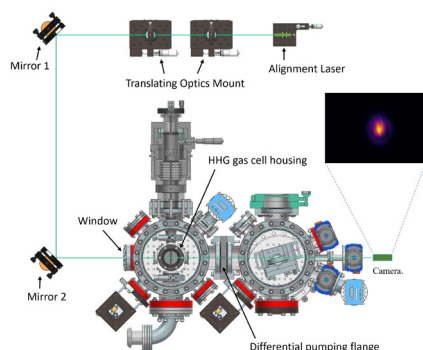


Figure 4: Schematic of the alignment setup for the in-vacuum sub-chamber.

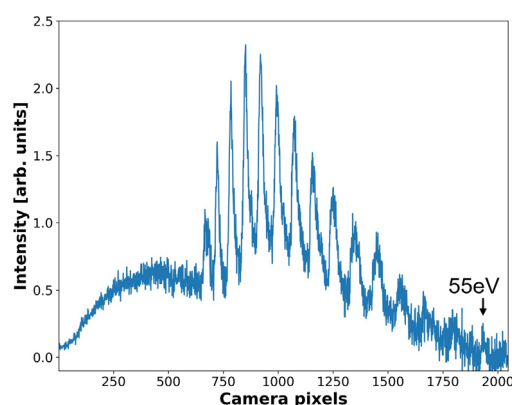


Figure 6: First HHG spectrum in argon (Ar).

HHG TESTBED

An HHG testbed has been developed and installed in the SXP experimental hutch to evaluate the performance of the complete HHG system under various operating conditions prior to final integration.

A top view of the HHG testbed is shown in Fig. 5. A custom-designed flat-field spectrometer (HP Spectroscopy – EVEREST) is integrated into the system to characterize the generated XUV beam.

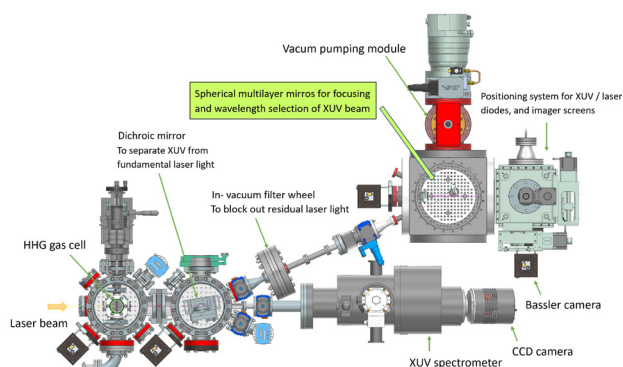


Figure 5: Top view of the HHG testbed setup.

The HHG testbed includes a vacuum cube and an integrated multilayer mirror system for focusing and selecting the desired XUV wavelength. The selected photon pulse train can be characterized using a photodiode mounted on the alignment manipulator stages.

Figure 6 shows the first HHG spectrum in argon, recorded at a backing pressure of 50 mbar. The spectrum was generated using 2 mJ pulses (50 fs / 1030 nm) at a repetition rate of 1.1 MHz (10 Hz burst mode), driven by the custom SASE3 pump-probe laser. Under these conditions, a cutoff photon energy of 55 eV was achieved. A pressure of 3×10^{-5} mbar was maintained inside the HHG source chamber, while the rejector chamber operated at a pressure on the order of 10^{-6} mbar, confirming the effective performance of the differential pumping scheme.

CONCLUSION

This contribution presents the main concept of mechanical design, key technological developments, implementation, and current status of experimental realization of the HHG source at the SXP Instrument.

The HHG source is a combination of ultra-high vacuum and optical components integrated into the SXP experimental hutch.

The HHG source in the SXP beamline is currently under testing and first stage of commissioning in real operational conditions. First spectra of generated HHG beam in the XUV spectrometer was measured.

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